

Training exercises

Database Normalization

Exercises

For each of the exercises, apply the appropriate normal forms, transforming the schemes into sets of schemes that are in the 1NF, 2NF and 3NF, justifying the normalization according to the marked functional dependencies. Also indicate any assumptions you need to make.

NOTE: Repeating groups are represented in curly brackets.

1. Medication form

Medication Form							
Patient number:		P1003		Department code:		Ort01	
Patient name:		John Smith		Department name:		Orthopedics	
Medication code	Medication name	Description	Dosage mg/ml	Administration method	Units per day	Start date	End date
10223	Morfina	analgesic	10	Oral	50	24-03-16	24-04-16
10351	Tetracycline	antibiotic	0.5	IV	10	24-03-16	17-04-16
10223	Morfina	analgesic	10	Oral	10	25-04-16	28-04-16

2. **Employee** (employee_num, employee_name, department_num, department_name, manager_num, manager_name, project_num, project_name, project_start_day, hours_worked).

3. **Purchase Order** (purchase_order_code, issue_date, supplier_code, supplier_address, {material_code, material_desc, purchased_qty, unit_value, total_item_value}, total_order_value).

4. **Invoice Table** (invoice_num, series, issue_date, customer_cd, customer_name, customer_address, customer_tin, product_code, product_desc, quantity_sold, price_sale, product_sale_total and general_invoice_total).

Assume that each note can have more than one product. Also suppose that the same products can be sold at different prices on different invoices.

What change would there be in terms of functional dependencies (FDs) if the sales price of a product were the same for all invoices?

5. **Patient** (patient_num, patient_name, room_num, room_description, room_num_beds, doctor_code, doctor_name, doctor_phone).

6. Consider the document shown in the following figure. Determine the set of relations in 3NF that represents the invoice information. Record all the assumptions you must make. Make any changes you deem necessary to the attribute set for efficient implementation. For each normal form, present the set of functional dependencies.



Inv# 06888

DATE: DECEMBER 19, 2011

TO Care Plus Clinic
3180 Collin Pl
P O Box 10 017
Wellington
NEW ZEALAND

SHIP
TO Salina Abgulmante
c/o Delivery Desk
90 Bear North Rd
Wellington Central 6011

SALESPERSON	SHIPPING METHOD	SHIPPING TERMS	DELIVERY DATE	PAYMENT TERMS	DUE DATE
Jene Pawjuk	UPS	Due on receipt	01/07/2011	by Credit Card	03/04/2012

QTY	ITEM #	DESCRIPTION	UNIT PRICE	DISCOUNT	LINE TOTAL
1	LP150-C	PS150-Centric SJ5_8, Lorentz Pump/Controller System.	\$268.00		\$268.00
1000	VD90a	AXC200 syringes	\$2.99		\$2990
3	TYDE3	JIN 2l reagent	\$98.00		\$294.00
1		Overnight shipping & icepack packaging	\$238.00		\$238.00
TOTAL DISCOUNT				6%	\$227.40
				SUBTOTAL	\$3562.60
				GST	\$534.39
				TOTAL	\$4096.99